

ALD²: Adaptive Layer-wise Denoising Decoding for Hallucinations Mitigation in Large Vision-Language Models

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ABSTRACT

Large vision-language models (LVLMs) have shown strong reasoning capabilities by leveraging Large language models (LLMs). However, hallucination remains a challenge for many LVLMs. Through our experiments, we identify that existing noise in shallow layers often leads to unstable token predictions during reasoning. We find that visual token pruning technologies along with multiplicative decoding effectively mitigate this interference. Building on this insight, we propose Adaptive Layer-wise Denoising Decoding (ALD²), a novel framework that mitigates hallucinations within LVLMs by dynamically selecting an optimal shallow layer for visual-token pruning and multiplicatively fusing the denoised distributions with that of the final layer. Extensive experiments on several benchmarks (CHAIR with 500 images; POPE with 500 images; MME with 2,374 cases) demonstrate that ALD² consistently improves the accuracy across multiple LVLMs. For example, we achieve an average accuracy improvement of 5.61% on POPE metrics using the InstructBlip-Vicuna-7B model and an increase of 88.56 MME score points in the MME evaluation benchmark with the LLaVA-Next-8B model.

1. Introduction

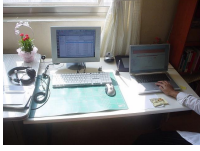
Building upon the rapid progress of Large language models (LLMs) (Yang et al., 2024; Zhao et al., 2024; Guo et al., 2025), recent Large vision-language models (LVLMs) (Liu et al., 2024a; Dai et al., 2023; Li et al., 2024) have achieved strong results in a wide range of vision-language tasks. However, multimodal fusion remains a core technical bottleneck, as models must align visual representations with linguistic semantics across substantial modality gaps (Wu et al., 2024; Li et al., 2026; Lee & Lee, 2026). Despite recent progress, visual hallucinations persist as a major challenge, reflecting the difficulty of achieving reliable visual grounding in complex scenarios. Mitigating these hallucinations is therefore crucial for building more robust and trustworthy multimodal systems.

To address this problem, recent works followed one mainstream method to mitigate hallucinations named Contrast Decoding (CD) (Leng et al., 2024; Wang et al., 2024b) that contrasting the LVLMs' output distributions under perturbed inputs to suppress ungrounded, biased predictions. Specifically, the Visual Contrastive Decoding (VCD) (Leng et al., 2024) method uses the image as a positive anchor to construct visually grounded positive and negative distributions and contrastively amplify tokens that align with the visual input. The Instruction Contrastive Decoding (ICD) (Wang et al., 2024b) method creates a negative distribution by performing a second forward pass with the negative instruction, then subtracts these "instruction-blind" logits from the full prompt logits to amplify tokens that truly depend on the given instruction. Despite their effectiveness in alleviating hallucinations, the improvements of these methods are still limited. Given that the LLM decoder is frequently employed in LVLMs as well as inspired by ALW (Zhou et al., 2025b), different from previous works, we view hallucinations from the perspective of noise signals within LVLMs. Similarly to LLMs, LVLMs also exhibit quite significant noise phenomena at shallow layers.

In LLMs, previous work (Zhou et al., 2025b) directly subtracting the prediction distribution of the shallow layer from that of the final layer can weaken the noisy context and produce the correct answer. Consequently, we first apply the same method to LVLMs but gain a slight effect. In order to analyze the reason, we conduct the experiments, and Figure 1 illustrates the underlying cause that correct tokens tend to receive higher probabilities than incorrect ones in shallow layers, yet the final layers of LVLMs still generate false tokens. This observation is consistent with the findings reported in (Wang et al., 2024a). This insight leads us to question whether the seemingly correct predictions in shallow layers are actually masked by substantial noise that becomes amplified during deep decoding, ultimately overriding the correct early cues and causing the model to output false tokens, thereby triggering hallucinations. Some recent works (Cao et al., 2023; Shang et al., 2025a; Zhang et al., 2026) pointed out that visual information is usually redundant during the inference of LVLMs. In view of this, we attempt to denoise by pruning a small part of redundant

ORCID(s):

Input Image:



Question:

Is there a chair in the image?

Answer:

No.

LVM response:

Yes, there is a chair in the image.

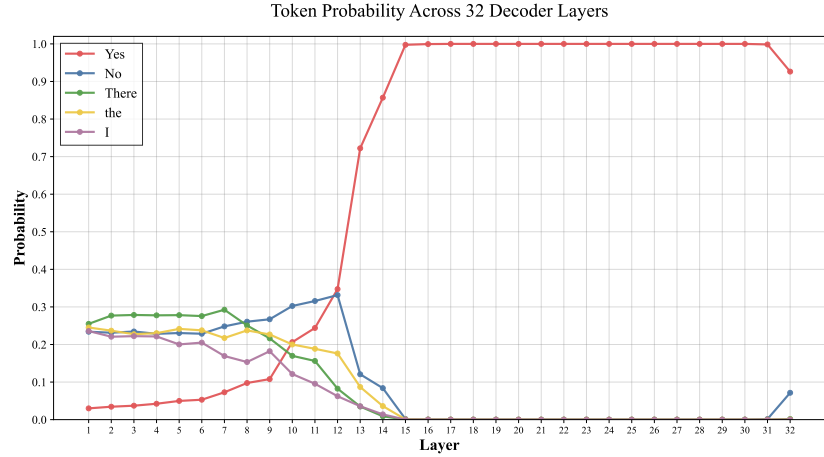


Figure 1: Token probability distribution across the 32 decoder layers of InstructBLIP-Vicuna-7B. For clearer visualization, we report the model's first predicted token along with the probabilities of the top-5 candidate tokens in the final layer.

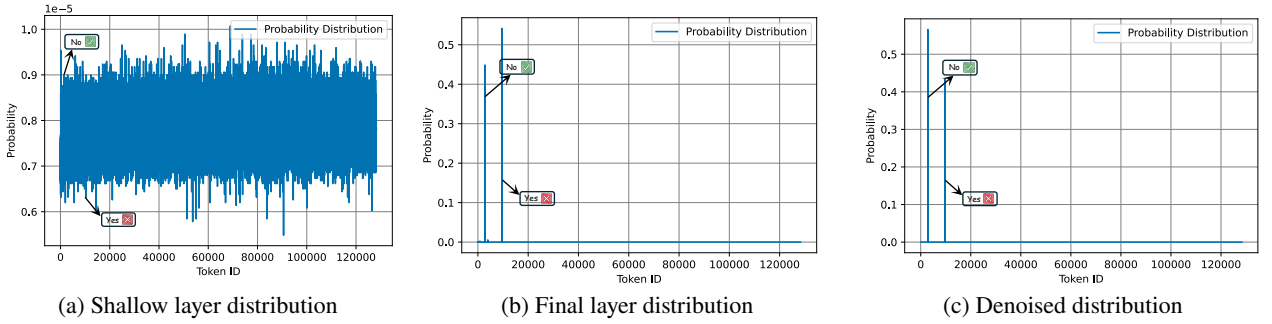


Figure 2: Illustration of the probability distributions during the decoding of InstructBLIP-Vicuna-7B. The prompt here is the same as Figure 1.

visual tokens on one of the shallow layers. Next, we multiply the final prediction distribution with that after pruning so that the probability of correct tokens will be amplified when simultaneously obtaining high-confidence support from the final layer and denoising-based reinforcement, thus preserving deep-layer syntactic reliability while incorporating pruned denoising gains. Figure 2 shows that after denoising, LVM could generate the correct answer “No”, and it preliminarily confirms that this method could work.

Therefore, we mitigate noise at the layer level and propose Adaptive Layer-wise Denoising Decoding (ALD²), a framework designed to mitigate hallucinations by pruning redundant shallow-layer visual tokens and multiplying their denoised distribution with the original one to amplify the denoising effect. To achieve this, we develop a quantification strategy to determine the optimal shallow layer for noise suppression. Furthermore, we introduce a lightweight language model (LM) that learns which layer of the LVMs requires noise suppression in an end-to-end manner and dynamically adjusts the multiplicative decoding strategy during the inference stage.

Specifically, we first identify the optimal noisy layer that requires the most suppression for different inputs and tasks. Then, an LM-based predictor is employed to capture the mapping between the prompt and the optimal denoised and multiplicative layer. During inference, the prompt is processed by both the LVM and the predictor. The LVM utilizes an optimal shallow layer selected by the predictor to generate the denoised prediction distribution and multiplies it with that of the final layer, effectively filtering noise from shallow layers while preserving key signals in the final layer. Extensive experiments demonstrate that ALD² achieves stable performance improvements and significantly mitigates

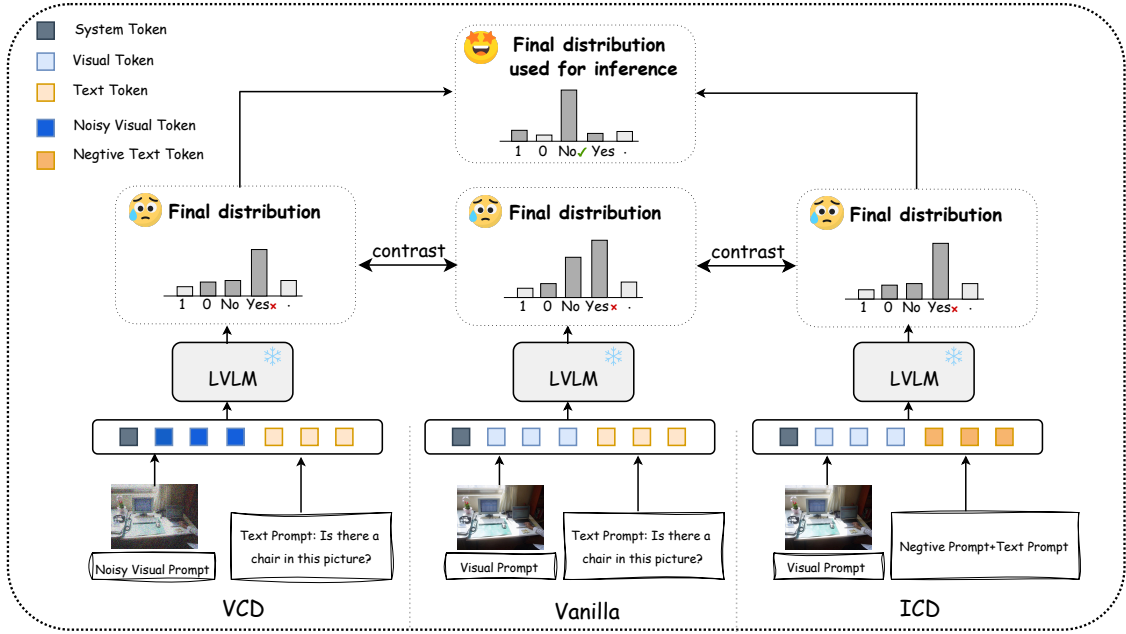


Figure 4: The current methods, VCD and ICD, to mitigate hallucinations in LVLMs

layer. Pruning removes redundant visual information, suppresses shallow-layer noise, and yields a probability distribution more concentrated on the target token.

This aligns with the observation in Figure 2, where we inspect the distribution immediately before decoding the final answer. In the final layer, the model hesitates between “yes” and “no” because their probabilities are nearly identical. This ambiguity originates from shallow-layer noise, where the probability of “yes” is much higher than that of “no”, injecting uncertainty into the final-layer prediction. After denoising, the model can confidently select the correct token.

2.2. Visual Token Pruning

As mentioned earlier in the paper, the shallow layers’ noise of LVLMs is relatively high, which remains a problem that cannot be ignored, and we find that noise cannot be eliminated by directly subtracting the log probabilities of different layers.

Therefore, according to some recent works (Cao et al., 2023; Shang et al., 2025a) that have pointed out that visual information is usually redundant during the inference process of LVLMs, in order to enable the model to focus more on valid visual information, redundant tokens are typically pruned and merged. Through our experiments, we observe that after pruning a small portion of visual tokens, the probability of correct tokens in the token probability distribution would be higher, while the probability of incorrect tokens would decrease, and that means the noise can be decreased in this way. Consequently, in LVLMs, we denoise the shallow layer through visual token pruning technologies.

Formally, for the decoder block, the text prompt, visual prompt, and generated tokens are concatenated and projected into Q vector, K vector, and V vector with three linear transformations W_q , W_k , W_v . During the calculation process of the self-attention mechanism, it A represents the attention score matrix before calculating the softmax function and multiplying by the V matrix. n represents the number of all tokens. Based on the attention score matrix A , we calculate the average attention score of each visual token $Score_{v_i}$ as the degree of attention the i -th visual token v_i receives. The attention score matrix A and visual token score $Score_{v_i}$ are calculated as follows:

$$A = \frac{QK^T}{\sqrt{d_k}} + M,$$

$$Score_{v_i} = \frac{1}{n} \sum_{j=1}^n A[j][i]. \quad (1)$$

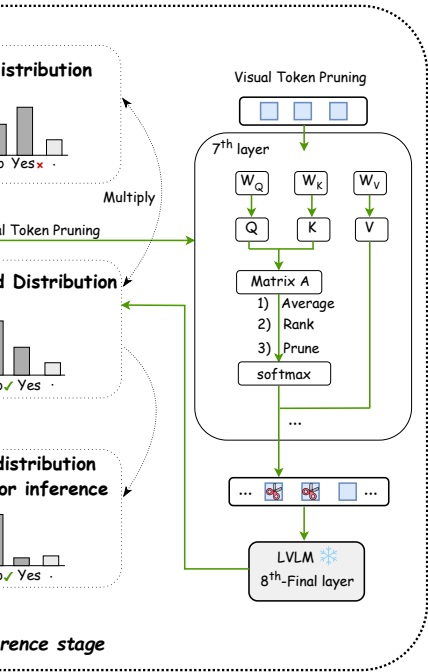


Figure 5: The workflow illustration of ALD² in the quantification and inference stage.

After calculating the score of all visual tokens, we rank them based on these average scores and set the hidden states' value of the tokens ranked in the bottom ten percent of average attention score to 0, indicating that these redundant visual tokens will be pruned. Lastly, the self-attention score is calculated as follows:

$$\text{Attention}(Q, K, V) = \text{softmax}(\text{Prune}(A) + M) V, \quad (2)$$

where d_k represents the dimension of Q , K , V . M represents the casual mask. *Prune* represents the visual token pruning technologies.

3. Method

3.1. Multiplicative Decoding

Typically, an LVLM’s decoder comprises an embedding layer, followed by M decoder layers indexed from 1 to M (embedding layer as 0), and ends with an output layer W_o . Given a prompt of length n , denoted by $X = \{x_0, x_1, \dots, x_{n-1}\}$, which consisted of visual tokens and text tokens, h^e represents the hidden states produced by the e -th layer. For each early layer $e \in [0, M]$, the probability distribution is then obtained as follows:

$$p_e(X) = \text{softmax}(W_o(h^e)). \quad (3)$$

To cancel LLMs’ noise, Zhou et al. (2025b) adopt the Self-Contrast Decoding method that contrasts the prediction of an early layer e with that of the final layer M . Specifically, they subtract the log probabilities of two layers. However, in LVLM, although noise still exists in shallow layers, the probability of correct tokens is often higher in the shallow layers, and the probability of correct tokens will gradually decrease layer by layer as shown in Figure 1. Therefore, the probability of generating correct tokens in the last layer is lower than that in the shallow layers, and if the Self-Contrast Decoding method is continued to be used, after subtracting the log probabilities of these two layers, the log probability of the correct token will be lower so that the decoding result will be worse. Consequently, we first cancel the noise of the early layer e through pruning visual tokens (§2.2). *Layer* represents the decoding layer of LVLM. The probability distribution is then calculated as follows:

$$\tilde{h}^e = Prune(h^e),$$

Algorithm 1 ALD² Decoding Strategy**Input:** Prompt $X = \{x_0, x_1, \dots, x_{n-1}\}$, LVLM, Layer, LM, Output Layer W_o , database D_b **Output:** Next token x_n

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1: Hidden states  $\{h^e\}_{e=0}^M \leftarrow \text{ForwardLVLM}(X)$ 
2:  $l \leftarrow \text{ForwardLM}(X)$ 
3: if  $l = -1$  then
4:    $\psi(x) = W_o(h^M)$ 
5: else
6:   Final_prob  $p_M = \text{softmax}(W_o(h^M))$ 
7:   Hidden state  $\tilde{h}^l = \text{Prune}(h^l)$ 
8:   for  $e = l$  to  $M$  do
9:      $\tilde{h}^{e+1} \leftarrow \text{Layer}_e(\tilde{h}^e)$ 
10:  end for
11:  Early_Prune_prob  $\tilde{p}_M = \text{softmax}(W_o(\tilde{h}^M))$ 
12:   $\psi(x) = \log(p_M \circ \tilde{p}_M)$ 
13:   $Mask \leftarrow$  Token indices not in  $\mathcal{V}_{head}$ 
14:   $\psi(Mask) \leftarrow -\infty$ 
15: end if
16: return  $x_n = \arg \max \text{softmax}(\psi(x))$ 

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$$\begin{aligned} \tilde{h}^M &= \text{Layer}_{e+1 \rightarrow M}(\tilde{h}^e), \\ \tilde{p}_M(X) &= \text{softmax}(W_o(\tilde{h}^M)). \end{aligned} \quad (4)$$

However, we do not use the probability distribution after pruning visual tokens for reasoning directly. To amplify the effect of denoising, we propose the Multiplicative Decoding method, where after pruning redundant visual tokens, the probability distribution obtained can be directly multiplied to the probability distribution of the final layer to effectively eliminate noise and ensure the accuracy of reasoning at the same time. Specifically, we add two of the log probabilities, yielding the following distribution:

$$\psi_e(x_n|X) = \log(p_M(x_n|X) \circ \tilde{p}_M(x_n|X)), \quad (5)$$

where $\circ$ denotes multiplication of corresponding entries in the probability distributions produced by the model.

For the actual selection of the early layer e , we first apply a quantification strategy on the training set. This allows us to identify the optimal layer and construct the corresponding prompt-layer data pairs (§3.2). Subsequently, we train a lightweight LM end-to-end (§3.3), which assists the LVLM in determining the most suitable optimal layer during inference (§3.4).

3.2. Quantification Strategy

In visual question answering (VQA) tasks, given a prompt X and a reference answer $A = \{a_1, a_2, \dots, a_m\}$ of length m , we treat each target token a_i independently. For every a_i , we identify the layer l that assigns the highest probability to this token, conditioned on X and its preceding tokens $a_{<i}$. The selection process is formulated as:

$$l = \arg \max_e \psi_e(a_i | X + a_{<i}), \quad e \in [0, M // 4 \times 3]. \quad (6)$$

If the final-layer probability already exceeds that of the selected layer, i.e., $p_M \geq \text{softmax}(\psi_l)$, denoising becomes unnecessary, and we assign $l = -1$. To reduce the search space, we restrict the candidate layers to the first three quarters of the model depth. For example, in LLaVA-1.5-7B with 32 layers, l ranges from -1 to 24. Figure 5 depicts the workflow for estimating the optimal layer. Given an answer of length m , this procedure yields m layer-prompt pairs.

3.3. Layer Prediction

As described above, for each task we obtain multiple $\langle X, l \rangle$ pairs. All pairs are collected into a database $D_b = \{\langle X_i, l_i \rangle\}_{i=0}^{N-1}$, where N denotes the total number of samples. We formulate layer prediction as a classification task and train a lightweight encoder (e.g., RoBERTa) to directly predict l from the input X .

To address class imbalance, each category is assigned a weight determined by its inverse frequency. Let p_θ denote the predicted distribution. The training objective is:

$$\mathcal{L} = -\frac{1}{b} \sum_{i=1}^b \sum_j w_j \cdot l_{ij} \log p_\theta(y_{ij} | X_i),$$

$$w_j = \frac{1/n_j}{\sum_j 1/n_j} \propto \frac{1}{n_j}, \quad (7)$$

where b is the batch size, and n_j is the number of samples belonging to the category j in $\mathcal{D}_b$. Example label distributions are provided in Appendix D.

3.4. Decoding Strategy

Inspired by prior work (Li et al., 2023a), which employs adaptive plausibility constraints to address contrastive false positives and false negatives, we adopt a similar filtering mechanism in our decoding process. Specifically, we retain tokens whose final-layer probabilities fall within a high-confidence region:

$$\{x_t \in \mathcal{V} : p_M(x_n | x_{<n}) \geq \alpha \max p_M(x_n | x_{<n})\}, \quad (8)$$

where $\alpha \in [0, 1]$ controls the truncation threshold. This step ensures that only tokens supported by the final layer remain valid candidates.

During inference, if a token obtains a high multiplicative probability but does not appear in the high-confidence set $\mathcal{V}_{\text{head}}$, it is removed from the decoding process. The resulting ALD² decoding score becomes:

$$\psi(x_n | X) = \begin{cases} \log(p_M(x_n | X) \circ \tilde{p}_M(x_n | X)), & x_n \in \mathcal{V}_{\text{head}}, \\ -\infty, & \text{otherwise,} \end{cases} \quad (9)$$

$$l = \arg \max_k p_\theta(y_k | X),$$

where l denotes the optimal predicted denoising layer. The full decoding procedure is detailed in Algorithm 1.

4. Experiments

4.1. Experimental Setup

Tasks. Our framework is evaluated across various tasks. We assess object hallucinations on **POPE** (Li et al., 2023c) and **CHAIR** (Rohrbach et al., 2018) metrics. As for CHAIR metrics, we randomly select 500 images from the MSCOCO 2017 (Lin et al., 2014) validation dataset. For the POPE metrics, we select 500 images from the MSCOCO 2014 (Lin et al., 2014) validation dataset and assign six queries to each image. For LVLMS' general ability, we employ **MME** (Fu et al., 2025) benchmark. These benchmarks are respected in the field for their diversity and breadth of tasks, allowing us to comprehensively analyze the performance of LVLMS from multiple perspectives. All these tasks are designed as open-ended generation tasks.

Models and Baselines. We employ three representative LVLMS: **LLaVA-1.5** (Liu et al., 2024a), **InstructBLIP** (Dai et al., 2023) at 7B scale, and **LLaVA-Next** (Li et al., 2024), **InternVL-3.5** (Wang et al., 2025) at 8B scale. We test our framework against four baselines: Greedy Decoding, Decoding by Contrasting Layers (**DoLa**) (Chuang et al., 2023), Instruction Contrastive Decoding (**ICD**) (Wang et al., 2024b), Visual Contrastive Decoding (**VCD**) (Leng et al., 2024). Greedy decoding selects the token with the highest probability at each step. DoLa leverages the Jensen-Shannon divergence between transformer layers within language models to enhance factuality. Notably, VCD, ICD, DoLa, and greedy decoding are all fully training-free methods, whereas ALD² introduces a lightweight classifier trained at the quantification stage to enable adaptive layer-wise denoising during decoding.

Details. In our experiments, the parameter α in Equation 8 is set to 0.1, following prior work (Li et al., 2023a). Consistent with Zhou et al. (2025b), we adopt a RoBERTa-base model (Liu et al., 2019) as the predictor. For open-ended generation tasks, Chain-of-Thought prompting (Wei et al., 2022; Yin & Wang, 2025) is applied when evaluating POPE and MME, while CHAIR is evaluated using direct prompting. Across all tasks and methods, the maximum number of generated tokens is set to 512 for fair comparison.

4.2. Results

4.2.1. Object Hallucinations

We evaluate hallucinations through giving LVLMs the text prompt: ‘Please help me describe the image in detail.’. Detailed CHAIR metrics are evaluated at the sentence level (C_S) and instance level (C_I). As shown in Table 1, our method outperforms the other four methods across three LVLMs. It is worth noting that VCD and ICD methods get worse results in comparison to the greedy decoding method, representing the limited effect of mitigating hallucinations.

The queries of POPE metrics are categorized into *Random*, *Popular*, and *Adversarial* types, following the standard POPE evaluation protocol. To enable the training of our classifier, we randomly split the dataset within each query type, using 80% of the samples for constructing the training set and the remaining 20% for building the test set. We then assess hallucination behaviors by prompting LVLMs with the following text query: ‘Is there a <object> in the image?’, which serves as the core probe for detecting object-related hallucinations. Results are illustrated in Table 1 and Table 2. As shown in Table 1, our method significantly reduces hallucination errors and achieves the highest average accuracy across all evaluated LVLMs. Notably, InstructBLIP-Vicuna-7B outperforms all baseline methods under every metric, validating the effectiveness of our approach. Following ALD², we retain the DoLa layer search space within $[0, M/4 \times 3]$ to ensure methodological consistency.

Table 1

Overall performance on CHAIR metrics across multiple LVLMs. **Bold** numbers indicate the best results in each column.

Method	LLaVA-1.5		InstructBLIP		LLaVA-Next		InternVL-3.5	
	$C_S \downarrow$	$C_I \downarrow$	$C_S \downarrow$	$C_I \downarrow$	$C_S \downarrow$	$C_I \downarrow$	$C_S \downarrow$	$C_I \downarrow$
Greedy	51.0	14.4	42.6	11.8	33.2	7.7	32.0	9.7
DoLa	47.8	14.6	43.0	11.7	35.6	8.8	30.8	10.1
ICD	47.2	13.8	55.6	17.5	39.2	10.1	39.4	11.2
VCD	50.8	13.5	50.2	14.2	47.2	10.7	40.4	10.3
ALD ² (Ours)	45.0	13.0	42.2	11.5	26.8	6.6	28.2	8.1

Table 2

Overall performance on POPE metrics across multiple LVLMs. **Bold** numbers indicate the best results in each column.

Model	Method	<i>Random</i>		<i>Popular</i>		<i>Adversarial</i>		<i>Average</i>	
		F1↑	Acc.↑	F1↑	Acc.↑	F1↑	Acc.↑	F1↑	Acc.↑
LLaVA-1.5-7B	Greedy	79.40	79.50	77.20	76.67	76.58	75.83	77.73	77.33
	DoLa	78.35	78.16	76.33	75.50	76.21	75.33	76.93	76.33
	ICD	83.79	84.33	82.05	82.50	79.20	79.17	81.68	82.00
	VCD	84.83	85.33	79.44	80.50	77.40	78.00	80.56	81.28
	ALD ² (Ours)	83.24	84.83	81.74	83.17	79.64	80.83	81.54	82.94
InstructBLIP-Vicuna-7B	Greedy	83.33	83.33	76.34	75.00	76.52	74.00	78.73	77.44
	DoLa	82.59	84.33	77.30	78.16	76.63	77.33	78.84	79.94
	ICD	83.97	84.67	79.80	79.67	77.91	78.83	80.56	81.06
	VCD	84.43	85.00	76.97	75.67	75.19	73.17	78.86	77.95
	ALD ² (Ours)	87.08	87.83	81.27	81.33	80.39	80.00	82.91	83.05
LLaVA-Next-8B	Greedy	86.87	87.50	84.07	84.33	82.26	82.17	84.40	84.67
	DoLa	86.87	87.50	84.07	84.33	82.26	82.17	84.40	84.67
	ICD	87.87	87.67	88.24	87.83	81.82	80.67	85.98	85.39
	VCD	87.37	87.67	83.70	83.83	82.87	82.50	84.65	84.67
	ALD ² (Ours)	90.00	90.33	89.76	89.50	85.57	85.33	88.44	88.39
InternVL-3.5-8B	Greedy	91.61	91.67	86.67	87.22	82.67	84.00	86.98	87.63
	DoLa	87.83	87.52	81.83	82.45	79.50	80.63	83.05	83.53
	ICD	91.25	91.33	85.95	85.83	83.64	82.33	86.94	86.50
	VCD	90.42	90.05	88.09	87.83	82.13	81.00	86.88	86.29
	ALD ² (Ours)	91.83	91.85	88.16	88.75	84.33	85.63	88.11	88.74

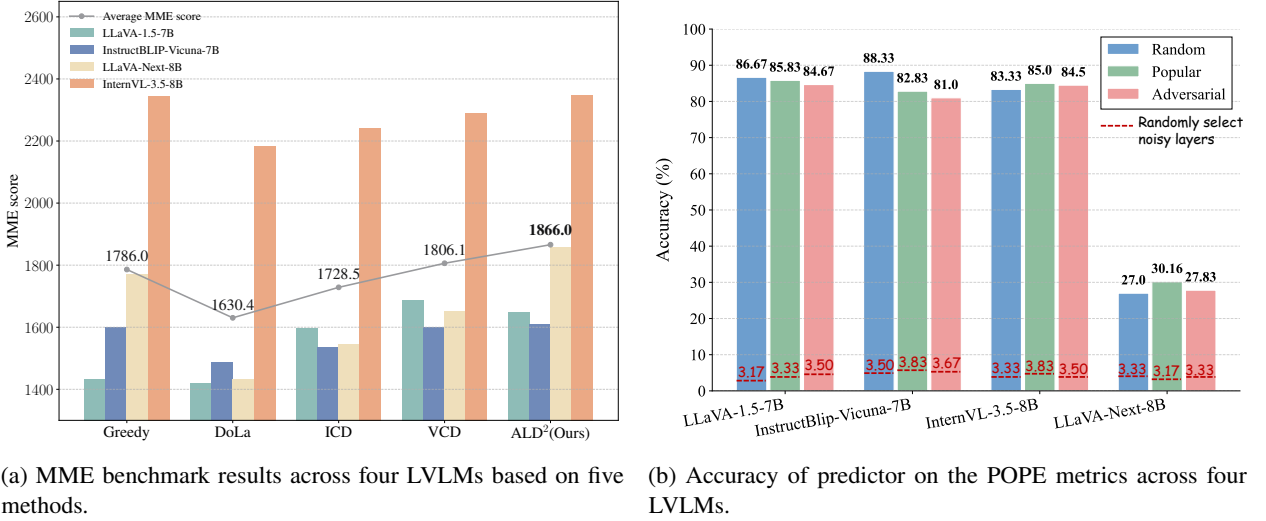


Figure 6: MME benchmark results (left) and accuracy of predictor (right).

4.2.2. Comprehensive Evaluation

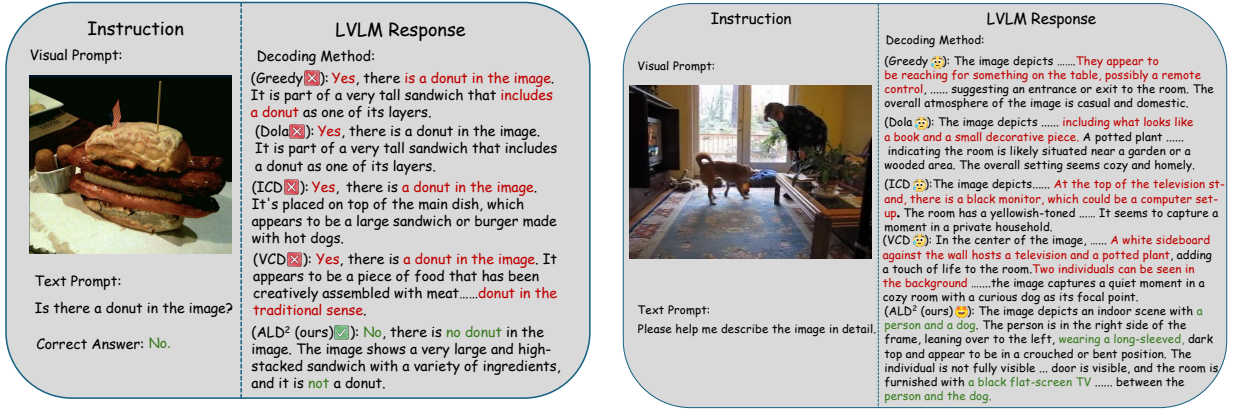
We verify the general ability of LVLMs on the MME benchmark (Fu et al., 2025), which comprises fourteen subtasks in total. Among them, four subtasks are designed to evaluate cognitive capabilities, including commonsense reasoning, numerical calculation, textual understanding, and code reasoning, while the remaining ten subtasks focus on perceptual abilities such as existence, count, position, color, poster recognition, celebrity identification, scene classification, landmark recognition, artwork understanding, and OCR. This comprehensive design enables MME to jointly assess both the semantic reasoning depth and the visual perception robustness of LVLMs, providing a balanced and multi-dimensional evaluation. Furthermore, the total number of cases in the MME benchmark is 2374. As illustrated in Figure 6a, our method ALD² achieves the strongest performance on the InstructBLIP-Vicuna-7B, InternVL-3.5-8B, and LLaVA-Next-8B models, and it also obtains the highest average MME score. These results demonstrate that ALD² not only mitigates hallucinations but also enhances general-purpose capabilities across diverse cognitive and perceptual tasks, confirming its effectiveness in improving overall LVLM reliability.

4.2.3. Accuracy of Predictor

For open-ended generation tasks, it is inherently challenging to determine whether a generated token is correct or incorrect at each step during inference, due to the absence of a definitive reference in the generation process. However, during training, we have access to the ground-truth sequences, allowing us to explicitly maximize the probability of the golden tokens after performing layer-wise multiplicative operations. This enables the model to learn which layers provide the most reliable information for reducing noise and producing accurate predictions.

To evaluate the effectiveness of our layer predictor, we report results on the POPE validation set, as shown in Figure 6b. For the LLaVA-1.5-7B, InstructBLIP-Vicuna-7B and InternVL-3.5-8B models, the predictor achieves high accuracy, with optimal layer selection exceeding 80% in most cases. This indicates that the predictor can reliably estimate the most informative layer for denoising given a specific prompt, thereby guiding the ALD² decoding process effectively. However, the LLaVA-Next-8B model exhibits lower prediction accuracy, around 30%, likely due to its more complex data distribution according to Appendix D and increased difficulty in distinguishing informative layers across diverse inputs.

Despite the relatively low accuracy of LLaVA-Next-8B in predicting the optimal layer, ALD² consistently surpasses baseline methods, demonstrating the robustness of our framework. Even when the predicted layer is not fully accurate, the denoising mechanism effectively improves generation quality by suppressing noise from shallow visual tokens and highlighting informative signals in deeper layers. Furthermore, as detailed in (§4.3), we conduct experiments where noisy layers are randomly selected based on our method. The results show that even randomly chosen layers outperform greedy decoding, and we additionally report the corresponding selection accuracy in Figure 6b. Notably, we fix the random seed to 42. Consequently, using a predictor is essential, though the improvement differs among LVLMs.



(a) One case for LLaVA-Next-8B based on five decoding methods when evaluating on POPE metrics. The red fonts represent the hallucination words, while the green fonts are the opposite. (b) One case for LLaVA-Next-8B based on five decoding methods when evaluating on CHAIR metrics. The red fonts represent the hallucination words, while the green fonts are the opposite.

Figure 7: Two inference cases of LLaVA-Next-8B based on ALD².

Furthermore, it can be observed that LLaVA-Next-8B achieves high performance gains despite low predictor accuracy. This discrepancy can be attributed to the objective used during the quantification stage. The layer predictor is trained to select the layer that maximizes the probability of the ground-truth token after layer-wise multiplicative fusion, rather than to directly predict token correctness. As a result, probability maximization does not necessarily guarantee correct token generation.

Additionally, according to Figure 9, a larger proportion of its layers exhibit high visual noise, making the separation between informative and noisy layers less distinct. Although the predictor achieves lower accuracy, correctly predicted cases almost always yield the desired token after denoising, and some cases predicted incorrectly even produce correct outputs due to effective suppression of shallow-layer noise.

In contrast, LLaVA-1.5-7B, InstructBLIP-Vicuna-7B and InternVL-3.5-8B show substantially reduced noise in deeper layers. Higher layer-selection accuracy does not always translate into correct token generation, as remaining errors are often dominated by semantic ambiguity rather than visual noise. These observations indicate that layer prediction accuracy and generation improvement are not strictly correlated.

4.3. Ablation

The effect of ALD². We explore why ALD² could mitigate hallucinations, owing to its success in the dynamic correction during inference of LVLMs. As shown in Figure 7a, merely our method generates the correct answers and gives the reasonable explanation. That is, after denoising the shallow layer and multiplicative decoding, we not only just convert the first token, such as “Yes” or “No”, but also explain why LVLMs convert the answer. For example, in Figure 7a, LVLM generates the complete answer ‘No, there is no donut in the image. The image shows a very large and high-stacked sandwich with a variety of ingredients, and it is not a donut.’ based on ALD², while others all generate the wrong answer. For more complex generation according to Figure 7b, only our method doesn’t generate any hallucinations. For example, when decoding based on the ICD method, LVLM generates the hallucinated words like “television setup”, “black monitor” and so on.

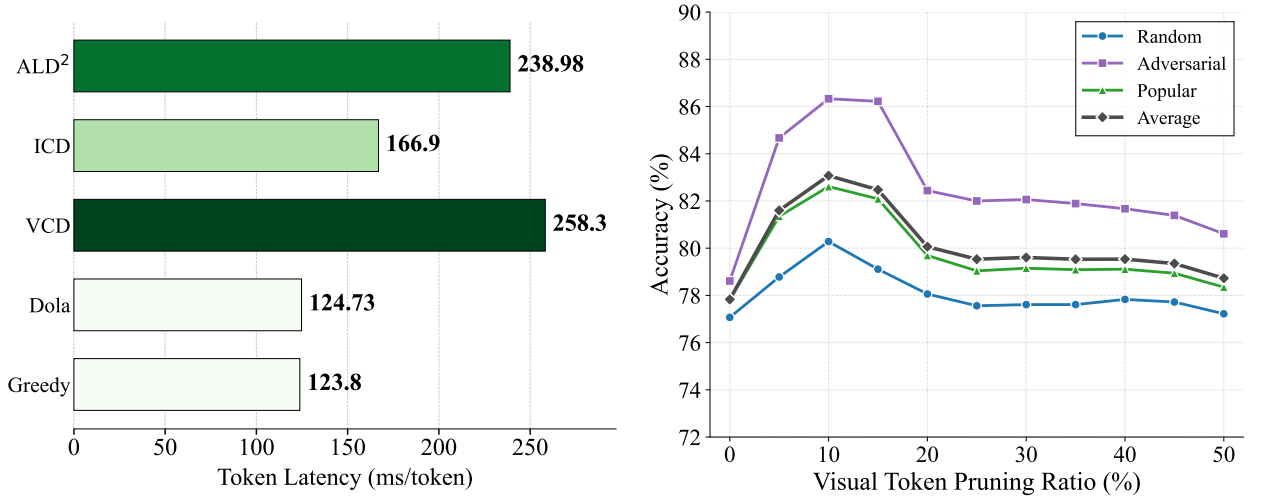
The proportion of visual tokens being pruned. In our approach, to obtain denoised probability distributions effectively, we prune a fixed 10% of visual tokens based on their average attention scores, as described in § 2.2. To evaluate the impact of different pruning proportions on inference accuracy, we conduct experiments varying the percentage of visual tokens removed. Figure 8b shows the average accuracy across three LVLMs LLaVa-1.5-7B, InstructBLIP-vicuna-7B, and LLaVA-Next-8B. For the average performance, pruning 10% of visual tokens achieves the highest average accuracy. As the pruning ratio increases beyond this point, the accuracy generally declines. These experiments are conducted using the POPE metrics. Notably, when no visual tokens are pruned (i.e., pruning ratio set to 0), the resulting accuracy is lower than that obtained with greedy decoding, and that also verifies the importance

of pruning redundant visual tokens. Furthermore, although we do not claim that a pruning ratio of 10% is universally optimal, our results indicate that pruning 10% of visual tokens is relatively effective.

Table 3

The importance of layer predictor. The numbers in parentheses indicate the change in accuracy when using random layer selection compared to using the predictor. The results in **bold** represent outcomes that exceed the greedy decoding results.

	<i>Random</i> ↑	<i>Popular</i> ↑	<i>Adversarial</i> ↑	CHAIR(C_S) ↓	CHAIR(C_I) ↓	MME score ↑
LLaVA-1.5-7B	81.17 (-3.66)	79.33 (-3.84)	77.50 (-3.33)	79.80 (+34.8)	34.50 (+21.5)	1116.15 (-531.35)
InstructBLIP-Vicuna-7B	85.83 (-2.00)	80.17 (-1.16)	78.50 (-1.50)	64.0 (+21.80)	18.7 (+7.20)	1593.33 (-16.86)
LLaVA-Next-8B	89.17 (-0.33)	87.17 (-2.33)	84.50 (-0.83)	30.40 (+3.60)	8.30 (+1.70)	1707.05 (-151.39)



(a) Averaged decoding latency per token in milliseconds (ms/token) for LLaVA-1.5-7B.

(b) The influence of the proportion of visual tokens pruned on the reasoning accuracy of the POPE metrics.

Figure 8: Comparison of latency (left) and accuracy of inference on POPE metrics (right).

Latency. We further examine the decoding latency. As illustrated in Figure 8a, ALD² registers a 94% increase compared to greedy decoding. Compared to DoLa (+0.75%), VCD(+109%) and ICD (+35%), although ALD² doesn't result in the most additional latency, the latency is still much higher than greedy decoding, and that means ALD² sacrifices some efficiency for the improvement of the accuracy. For example, for the LLaVA-1.5-7B model, ALD² gets a 5.61% improvement in average accuracy. Moreover, compared with greedy decoding, ALD² performs additional inference over one quarter to all layers after the main decoding step, which we identify as the primary source of the increased latency. Regarding scalability, while our evaluation focuses on standard-length text responses and moderate-resolution visual inputs, the relative latency overhead of ALD² may grow for longer responses or higher-resolution images. Since both the additional layer-wise inference and token processing scale with the number of tokens and visual features, scenarios with extended dialogues or dense visual inputs could experience further latency increases. Notably, the latency can be reduced by leveraging our visual token pruning strategy, where less informative tokens are directly removed from the computation graph rather than merely zeroing their hidden states. This reduces memory usage and computation in subsequent layers, offering a practical avenue to mitigate latency while retaining most of the accuracy improvements.

Effect of Multiplicative Decoding and Visual Token Pruning. To further understand why ALD² is effective, we compare the full ALD² framework with three variants: greedy decoding, pruning-only decoding, and a replacement of multiplicative fusion with additive fusion. As shown in Table 4, combining visual token pruning with multiplicative decoding consistently outperforms additive fusion, pruning-only decoding, and greedy decoding across random, popular, and adversarial settings. While pruning alone already improves performance over greedy decoding, indicating the effectiveness of suppressing redundant visual noise, its gains saturate without an explicit fusion mechanism. Replacing multiplication with addition leads to noticeable degradation, especially on popular and adversarial subsets, suggesting

that additive fusion fails to sufficiently suppress residual noisy signals. In contrast, multiplicative decoding better emphasizes tokens jointly supported by both the denoised shallow layer distribution and the final layer distribution, acting as a confidence-aware agreement mechanism. These results confirm that the superiority of ALD² arises from the joint design of visual token pruning and multiplicative decoding, rather than from either component in isolation.

Table 4

Ablation study on fusion strategy and visual token pruning on POPE based on LLaVA-Next-8B. **Bold** numbers indicate the best results in each column.

Method	<i>Random</i>		<i>Popular</i>		<i>Adversarial</i>		<i>Average</i>	
	F1↑	Acc.↑	F1↑	Acc.↑	F1↑	Acc.↑	F1↑	Acc.↑
Greedy	86.87	87.50	84.07	84.33	82.26	82.17	84.40	84.67
Prune Only	90.00	90.33	85.00	86.24	81.62	78.83	85.54	85.13
Add + Prune	89.48	89.50	85.35	84.67	82.46	81.00	85.76	85.06
ALD²(Multiply + Prune)	90.00	90.33	89.76	89.50	85.57	85.33	88.44	88.39

Random layer. We also verify the importance of the predictor by randomly selecting the probability distributions of the noisy layer and multiplying them with the probability distribution of the last layer. As reported in Table 3, the results clearly show that this random selection strategy causes a substantial decline in performance across all evaluation settings, indicating that blindly choosing layers disrupts the structured information necessary for generating reliable outputs. Nevertheless, it is noteworthy that even under random selection, more than 50% of the cases still outperform greedy decoding baselines (highlighted in **bold**), suggesting that leveraging intermediate layers inherently possesses a degree of robustness and diversity. This observation emphasizes that the predictor plays a crucial role in identifying layers that consistently provide beneficial signals, rather than relying on chance. Therefore, a predictor equipped with a principled quantification strategy is essential for learning the patterns of optimal layer occurrence, and the results further demonstrate that ALD² maintains effectiveness even when the selected layer is not the absolute optimal one.

5. Related work

5.1. Hallucinations Mitigation in LVLMS

Hallucination mitigation in LVLMS has attracted increasing attention due to the inherent difficulty of aligning visual and textual modalities. Prior studies mainly explore three directions (Huo et al., 2024). The first direction strengthens model robustness through instruction tuning or alignment with curated, **visually grounded datasets** (Lee et al., 2022; Gunjal et al., 2024; Liu et al., 2023; Jing et al., 2024, 2025; Liu et al., 2024c; Zhang et al., 2024), including approaches based on **factually augmented reinforcement learning and fine-grained feedback to improve visual faithfulness** (Jing & Du, 2024; Sun et al., 2024; Shang et al., 2025b). The second direction applies post-hoc correction or verification with auxiliary analysis modules to detect and **suppress hallucinated content after generation** (Manakul et al., 2023; Zhou et al., 2024a; Chang et al., 2024), **often relying on external reasoning paths, multi-view analysis, or additional verification signals**.

The third line of work adjusts the decoding process to mitigate hallucinations through contrastive, constraint-based, or uncertainty-aware strategies (Li et al., 2023b; Chuang et al., 2023; Liu et al., 2024b). Representative methods include visual contrastive decoding and its variants (e.g., VCD and Med-VCD) (Mahdavi et al., 2026; Leng et al., 2024), instruction contrastive decoding (ICD) (Wang et al., 2024b), dynamic correction decoding (DeCo) (Wang et al., 2024a), **multi-view multi-path reasoning** (Qu et al., 2025), and self-introspective decoding (SID) (Huo et al., 2024), which dynamically evaluates token importance during generation. These decoding-time approaches are lightweight and model-agnostic and have shown promising effectiveness in reducing hallucinations without retraining large models.

Despite their success, existing methods largely operate at the data level or the output distribution level and rarely examine noise patterns within the model itself, particularly the instability introduced by redundant visual tokens in shallow layers. Such shallow-layer noise can propagate through deeper layers during decoding and ultimately trigger hallucinated predictions. To address this overlooked factor, we propose ALD², a decoding-time method that explicitly targets shallow-layer noise through adaptive visual token pruning and confidence-aware layer-wise fusion, enabling more stable and visually grounded generation.

5.2. Early-Exiting Approaches

Panda et al. (2016) were among the first to explore early exiting in neural networks as a strategy to accelerate inference. Over the years, early exit mechanisms have become standard in efficient LLMs (Khoshnoodi et al., 2024). For instance, LayerSkip (Elhoushi et al., 2024) uses early exit layers for speculative decoding, while SkipDecode (Corro et al., 2023) defines a single exit point for each token across all sequence positions in a batch. Other contemporaneous methods, including EE-LLM (Chen et al., 2024) and ConsistentEE (Zeng et al., 2024), focus on extreme efficiency, often pruning up to 40% of layers while maintaining nearly full performance (Sajjad et al., 2023). In contrast, our approach treats early exit layers not as a speed-up mechanism but as an auxiliary tool for denoising. Through combining a shallow early layer with the final layer, we capture informative patterns that help mitigate hallucinations, emphasizing generation quality over computational efficiency.

5.3. Contrastive Decoding

Contrastive decoding has been extensively studied in recent years. Su et al. (2022) introduced Contrastive Search to enhance semantic distinctions among candidate tokens, while Li et al. (2023a) proposed Contrastive Decoding that exploits differences between expert and amateur models. Subsequent works (Gera et al., 2023; Chuang et al., 2023) explored layer-wise contrast within a single backbone model, improving text generation by leveraging intermediate representations. With the recent extension of LLM techniques to LVLMs, researchers (Leng et al., 2024; Wang et al., 2024b) have applied contrastive decoding to open-ended vision-language tasks, aiming to reduce hallucinations by contrasting token probabilities across layers. Inspired by these ideas, our method diverges from direct contrastive decoding. We propose a multiplicative decoding strategy that amplifies the effect of denoising while explicitly leveraging layer-wise noise patterns from shallow visual tokens. This allows LVLMs to suppress spurious signals and focus on informative content, effectively mitigating hallucinations without relying solely on conventional contrastive objectives.

6. Theoretical and Practical Implications

Our work provides both theoretical and practical insights into mitigating visual hallucinations in large vision-language models. From a theoretical perspective, our findings suggest that hallucinations are not solely determined by the final decoding layer but are strongly influenced by noisy representations in shallow layers. By empirically demonstrating that correct tokens often receive relatively higher probabilities at shallow layers before being overridden during deep decoding, this work reframes hallucination as a layer-wise noise propagation problem rather than a purely semantic misalignment issue. This perspective complements and extends existing contrastive decoding approaches, which primarily focus on input perturbations, by highlighting the importance of internal layer dynamics during inference.

From a practical perspective, the proposed ALD² framework offers a lightweight yet effective solution for reducing hallucinations in LVLMs, operating entirely at the decoding stage and easily integrating into existing pipelines. Its adaptive layer selection mechanism enables dynamic noise suppression tailored to different inputs, making it suitable for real-world deployment scenarios where reliability is critical. In practice, we believe the trade-off between increased decoding latency and improved trustworthiness is acceptable in settings where hallucination reduction and reasoning reliability are prioritized over strict efficiency constraints, such as offline evaluation, safety-critical decision support, medical or scientific analysis, and high-stakes human-in-the-loop applications. Conversely, for latency-sensitive scenarios like real-time interactive systems, greedy decoding or other lightweight strategies may be preferable. These considerations underscore that layer-wise denoising during decoding is both theoretically grounded and practically effective.

7. Conclusion

In this work, we investigate the role of attention noise in causing hallucinations within LVLMs, analyzing how noise propagates across different layers. We find that, similar to LLMs, the accumulation of noise in shallow layers can lead to unstable token predictions in later layers, often resulting in hallucinations. To address this challenge, we propose Adaptive Layer-wise Denoising Decoding (ALD²), a search-based decoding framework that dynamically identifies and suppresses noisy signals in shallow layers while preserving critical information in deeper layers. Our approach includes a quantification strategy to evaluate layer-wise contributions and a predictor for optimal layer selection. Extensive experiments show that ALD² effectively reduces hallucinations in LVLMs. Further analysis indicates that its success stems from selectively focusing on the most informative visual tokens, enhancing prediction reliability through targeted noise suppression.

Limitations

While the Adaptive Layer-wise Denoising Decoding (ALD²) method has significantly mitigated hallucinations and improved the reasoning reliability of Large Vision-Language Models by dynamically suppressing noise in shallow layers and emphasizing critical signals in deeper layers, it still has several limitations:

- ALD² denoises shallow layers by pruning a fixed proportion of visual tokens based on attention scores. This design choice favors simplicity and model-agnostic deployment, but it does not explicitly account for more complex vision-aware internal dynamics, such as adaptive token selection, vision-aware head behaviors, task-specific hallucination mitigation, or in-context visual reasoning (Zhou et al., 2025a; He et al., 2025; Zhou et al., 2024b). More precise token pruning strategies or dynamically adjusted pruning ratios may further improve denoising effectiveness. For example, Huo et al. (2024) proposed Self-Introspective Decoding (SID), which dynamically evaluates the importance of vision tokens during decoding to adaptively reduce hallucinations in large vision-language models.
- ALD² requires a training phase prior to inference. Although the predictor is lightweight, it still introduces additional computational overhead and latency. Our experiments demonstrate that this overhead is non-negligible, which may limit the applicability of ALD² in latency-sensitive scenarios.
- While ALD² mitigates hallucinations by reducing noise in shallow layers, it does not explicitly model hallucination mechanisms or perform hallucination detection or correction, particularly in long-form responses (Huang et al., 2025; Zheng et al., 2025). As a result, in certain cases where LVLMs produce highly confident but incorrect predictions, residual hallucination effects may still exist, despite being alleviated by ALD².

Data availability

The data and code will be released after the paper’s publication.

A. Details of Metric Calculations

CHAIR. CHAIR (Rohrbach et al., 2018) is a commonly adopted metric in image captioning for detecting object hallucinations. It determines whether hallucinated objects are present by comparing objects extracted from generated captions with ground-truth labels and reports performance at both the instance level (CHAIR_I) and the sentence level (CHAIR_S), as formalized as follows:

$$\begin{aligned} C_I &= \frac{|\{\text{hallucinated objects}\}|}{|\{\text{all mentioned objects}\}|}, \\ C_S &= \frac{|\{\text{captions with hallucinated objects}\}|}{|\{\text{all captions}\}|}. \end{aligned} \quad (10)$$

POPE. The POPE metrics (Li et al., 2023c) assess object-related hallucinations in large vision-language models (LVLMs) through targeted textual queries. For each image, queries ask about the presence of specific objects, enabling fine-grained evaluation. Performance is measured using **accuracy (Acc.)** and **F1 score**, which are defined as follows:

$$\begin{aligned} \text{Accuracy (Acc.)} &= \frac{\text{True Positives (TP)} + \text{True Negatives (TN)}}{\text{Total Questions}}, \\ \text{Precision} &= \frac{\text{TP}}{\text{TP} + \text{False Positives (FP)}}, \\ \text{Recall} &= \frac{\text{TP}}{\text{TP} + \text{False Negatives (FN)}}, \\ \text{F1 Score} &= \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \end{aligned} \quad (11)$$

Input image :



Question:

Please help me describe the image in detail.

LLM response:

The image depicts a room with a collection of various items.

	1	2	3	4	5	6	7	8	9	10	11	12	13
Denoised	0.25	0.00	0.00	0.00	0.06	0.00	0.03	0.03	0.00	0.33	0.00	0.00	0.18
32	-0.00	0.00	0.55	0.00	1.35	0.71	3.05	1.69	1.27	2.78	0.00	2.23	1.46
30	11.64	11.58	10.84	9.27	9.67	10.08	10.89	10.75	10.63	10.98	8.80	10.67	10.46
28	11.63	11.56	11.30	10.77	11.11	11.41	11.27	11.40	11.30	11.34	11.32	11.20	11.27
26	11.66	11.62	11.49	11.27	11.47	11.59	11.44	11.55	11.49	11.48	11.51	11.41	11.47
24	11.70	11.69	11.66	11.55	11.64	11.66	11.59	11.66	11.62	11.62	11.63	11.56	11.61
22	11.73	11.72	11.71	11.67	11.70	11.70	11.68	11.70	11.69	11.69	11.69	11.66	11.68
20	11.74	11.74	11.73	11.72	11.73	11.73	11.73	11.73	11.73	11.73	11.73	11.72	11.72
18	11.74	11.74	11.73	11.73	11.73	11.74	11.73	11.73	11.74	11.73	11.73	11.73	11.73
16	11.74	11.74	11.74	11.74	11.74	11.74	11.74	11.74	11.74	11.74	11.74	11.74	11.74
14	11.75	11.75	11.75	11.75	11.74	11.75	11.75	11.75	11.75	11.75	11.75	11.75	11.75
12	11.75	11.75	11.75	11.75	11.75	11.75	11.75	11.75	11.75	11.75	11.75	11.75	11.75
10	11.75	11.75	11.75	11.75	11.75	11.75	11.75	11.75	11.75	11.75	11.75	11.75	11.75
8	11.75	11.75	11.74	11.74	11.75	11.75	11.75	11.75	11.75	11.75	11.75	11.75	11.75
6	11.74	11.74	11.74	11.74	11.75	11.75	11.74	11.75	11.74	11.75	11.75	11.75	11.75
4	11.74	11.74	11.74	11.74	11.74	11.74	11.74	11.74	11.74	11.74	11.74	11.74	11.74
2	11.73	11.73	11.73	11.73	11.73	11.73	11.73	11.73	11.73	11.73	11.73	11.73	11.73
0	11.73	11.73	11.73	11.73	11.73	11.73	11.73	11.73	11.73	11.73	11.73	11.73	11.73
	'n'	'The'	'image'	'depicts'	'a'	'room'	'with'	'a'	'collection'	'of'	'various'	'items'	'!'

Figure 9: More preliminary cases (case 1, results on LLaVA-Next-8B).

Here, TP, TN, FP, and FN are counted based on whether the model correctly identifies the presence (“yes”) or absence (“no”) of objects in the images.

MME. The MME benchmark (Fu et al., 2025) is a comprehensive evaluation suite designed to measure both perception and cognition capabilities of multimodal large language models (MLLMs). It contains 14 tasks divided into two main categories:

- **Perception** (10 tasks): existence, count, position, color, posters, celebrity, scene, landmark, artwork, and OCR;
- **Cognition** (4 tasks): commonsense reasoning, numerical calculation, text translation, and code reasoning.

All questions in MME are yes/no questions. To assess consistency and robustness, the benchmark deliberately designs **two questions per image**.

The scoring system uses two key metrics:

- **Accuracy**: the percentage of correctly answered individual questions;
- **Accuracy+** (also called Acc+ or strict accuracy): the percentage of images where **both** questions are answered correctly.

For each task, the task score is computed as $100 \times (\text{Accuracy} + \text{Accuracy+})$, yielding a maximum of 200 points per task. The Perception score is the sum of the 10 perception task scores (max 2000), the Cognition score is the sum of the 4 cognition task scores (max 800), and the overall **MME Total Score** is the sum of the Perception and Cognition scores (max 2800).

B. More Preliminary Cases

We present additional preliminary cases in Figures 9 and 10. The observed behavior remains consistent with the phenomenon illustrated in Figure 3. Notably, the newly added cases exhibit patterns highly similar to those in Figure 3 across two additional LLMs, demonstrating that the noise issue is not model-specific but instead a widespread characteristic shared among LLMs.

Furthermore, the denoising operation consistently reduces prediction uncertainty in the final layer, leading to higher confidence in the correct token. This effect stems from suppressing residual noise inherited from shallow visual tokens, allowing the final-layer representation to focus more on semantically relevant information and reducing the dominance of hallucinated predictions. As a result, the decoding process becomes more stable and better aligned with the underlying visual evidence. Figure 2 provides a token-level view rather than a distribution-level one, enabling a fine-grained analysis of how individual token probabilities are reshaped after applying the denoising mechanism.

Input image :



Question:

Please help me describe the image in detail.

LVLM response:

In the image, a group of three elephants, two of which are young.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Denoised	0.44	0.37	-0.00	-0.00	-0.00	-0.00	-0.00	-0.00	-0.00	-0.00	-0.00	-0.00	-0.00	-0.00	-0.00	-0.00	-0.00	-0.00	-0.00	-0.00
32	0.46	1.47	1.44	1.38	0.59	0.20	0.01	1.19	2.08	0.01	1.19	0.77	0.00	0.01	1.44	1.68	1.39	0.62	0.51	2.61
30	0.94	0.87	0.02	1.36	0.19	0.17	0.00	0.81	0.36	0.00	0.36	0.18	0.03	0.00	0.80	0.99	0.32	0.55	0.01	2.13
28	0.03	0.34	0.45	0.14	0.70	0.00	0.00	0.77	1.89	0.00	0.00	0.01	0.62	0.01	0.36	0.62	1.31	0.65	0.14	1.35
26	0.02	2.66	2.03	1.04	0.37	0.01	0.02	0.88	1.23	0.00	0.01	0.02	0.37	0.31	1.28	0.93	1.62	0.21	0.69	0.37
24	0.82	9.14	9.62	8.67	0.31	0.03	0.08	0.75	2.31	0.00	0.05	0.12	0.59	4.50	2.30	1.02	0.71	0.41	0.94	0.63
22	5.53	10.00	10.01	9.71	1.45	0.18	1.65	0.45	4.27	0.22	0.17	0.29	2.08	7.82	4.89	1.78	0.76	0.72	0.69	0.54
20	9.54	10.15	10.14	10.11	4.29	3.60	8.89	3.80	6.83	8.38	5.71	6.45	8.43	9.45	9.33	3.95	8.67	3.76	7.93	9.00
18	10.02	10.23	10.21	10.19	9.88	10.02	9.84	9.84	10.05	9.80	10.02	10.07	10.09	10.02	10.02	9.17	10.03	8.52	10.04	10.05
16	10.16	10.27	10.27	10.26	10.18	10.17	10.10	10.17	10.20	10.10	10.19	10.20	10.20	10.17	10.17	10.14	10.16	10.16	10.18	10.18
14	10.29	10.31	10.31	10.32	10.29	10.29	10.25	10.28	10.29	10.25	10.28	10.27	10.29	10.23	10.27	10.25	10.26	10.26	10.24	10.26
12	10.33	10.34	10.34	10.34	10.33	10.33	10.30	10.32	10.32	10.30	10.32	10.31	10.31	10.29	10.30	10.32	10.31	10.30	10.30	10.30
10	10.35	10.35	10.34	10.35	10.34	10.34	10.33	10.34	10.34	10.33	10.34	10.34	10.34	10.32	10.33	10.34	10.34	10.33	10.31	10.33
8	10.36	10.36	10.36	10.36	10.36	10.35	10.34	10.35	10.35	10.34	10.34	10.35	10.35	10.34	10.34	10.35	10.34	10.34	10.34	10.34
6	10.37	10.37	10.36	10.36	10.36	10.36	10.36	10.36	10.36	10.36	10.36	10.36	10.36	10.35	10.36	10.36	10.36	10.36	10.35	10.36
4	10.37	10.37	10.37	10.37	10.37	10.37	10.37	10.37	10.37	10.37	10.37	10.37	10.37	10.37	10.36	10.37	10.37	10.37	10.37	10.37
2	10.38	10.38	10.38	10.38	10.38	10.38	10.38	10.38	10.38	10.38	10.37	10.38	10.38	10.37	10.38	10.38	10.38	10.38	10.38	10.38
0	10.38	10.38	10.38	10.38	10.38	10.38	10.38	10.38	10.38	10.38	10.38	10.38	10.38	10.38	10.38	10.38	10.38	10.38	10.38	10.38
l-th early exit layer	"un"	"un"	"In"	"the"	"image"	','	"a"	"group"	"of"	"three"	"ele"	"ph"	"ants"	','	"two"	"of"	"which"	"are"	"young"	','

Figure 10: More preliminary cases (case 2, results on LLaVA-1.5-7B).

C. More Experiment Details

Alignment of LM and LVLM. We believe that a perfect match between the RoBERTa tokenizer and the LVLM tokenizer is not necessary. When RoBERTa predicts the optimal denoised layer, its input is the full, raw text, rather than the ID sequence tokenized by LVLM. The task for RoBERTa is to predict which layer’s noise needs to be suppressed most for the current input, which is a sentence-level or paragraph-level semantic decision, rather than a fine-grained subword match. In fact, this approach is similar to the length prediction method used in LLM scheduling (Stojkovic et al., 2024), where BERT is employed to predict the possible output length for an LLM given the input to plan memory usage, and such methods often do not require alignment between PLM and LLM tokenizers.

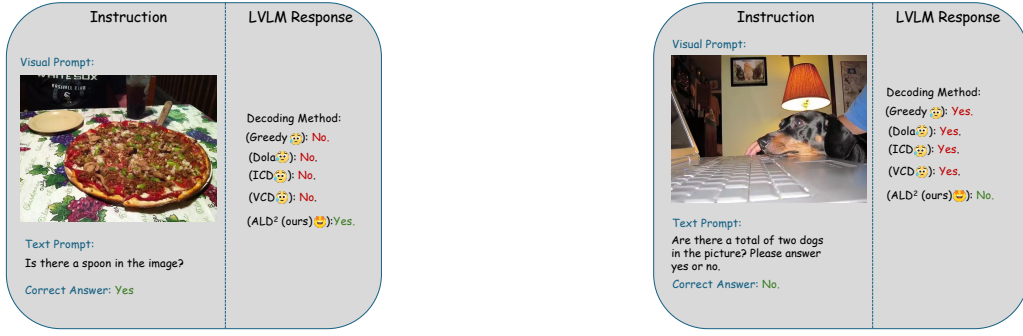
GPUs. The main experiments were conducted on an NVIDIA GeForce RTX 3090. When running ALD² with LLaVA-1.5-7B as the LVLM and RoBERTa-base as the predictor, one RTX 3090 is required. The predictor remains the same. Since the predictor’s inference speed is very fast and does not require a lot of GPU communication, the budget is essentially consistent with simply inferring the three LVLMs.

Others. The number of shots assigned for POPE metrics and the MME benchmark is one. Moreover, for the InstructBLIP-Vicuna-7B model, multi-image input is not supported, so the image is not contained, while for the LLaVA-Next-8B model and the LLaVA-1.5-7B model, the image is contained in the shot.

D. Most Frequent Layer

We analyze the distribution of the optimal denoised layer obtained when training on the three POPE subsets across the three LVLMs. The detailed statistics are reported in Table 5, Table 6, and Table 7. For both LLaVA-1.5-7B and LLaVA-Next-8B, we observe that the “no-denoised” option is selected most frequently, while a diverse range of other layers also appear. This phenomenon indicates that the necessity of denoising is highly input-dependent rather than fixed across samples. Notably, the LLaVA-Next-8B model exhibits a noticeably more complex distribution, and layer 2 becomes the dominant choice. This observation aligns well with our earlier analysis on predictor accuracy, where the predictor trained on LLaVA-Next-8B shows lower accuracy due to the more irregular and widely dispersed distribution. In this sense, the distribution itself provides a natural explanation for the predictor’s difficulty in capturing the optimal layer for this model.

Another consistent trend across all three LVLMs is that shallow layers appear more frequently than deeper layers. This pattern further confirms our hypothesis that early layers are more affected by noise from visual tokens and therefore benefit more from denoising interventions. Furthermore, the overall distribution of optimal layers supports our design choice of focusing on the first three quarters of the model depth. Since nearly all frequently selected layers fall within



(a) One case for InstructBLIP-Vicuna-7B based on five decoding methods when evaluating on POPE metrics. The **red** fonts represent the hallucination words, while the **green** fonts are the opposite.

(b) One case for LLaVA-1.5-7B based on five decoding methods when evaluating on the MME benchmark. The **red** fonts represent the hallucination words, while the **green** fonts are the opposite.

Figure 11: Two inference cases of the InstructBLIP-Vicuna-7B and LLaVA-1.5-7B models across five decoding methods.

this range, restricting our search space to these layers is empirically well justified. This concentration of effective layers also suggests that most meaningful denoising signals emerge before the deepest stages of the network, where representations tend to stabilize and additional suppression yields diminishing returns. Together, these observations indicate that early-stage representations play a critical role in controlling hallucinations and that targeted denoising at appropriate depths is more effective than uniformly applying denoising across all layers.

Table 5

The statistics of the optimal layer. We report the results on POPE metrics across three LVLMS. The layer requiring no contrast operation appears most frequently, while other layers are also involved. Moreover, shallow layers appear more frequently than higher layers. This figure shows the results of the *Random* subset on POPE metrics.

Layer	-1	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	20	21
LLaVA-1.5-7B	2176	12	7	0	2	0	0	164	1	0	1	0	0	0	4	13	0	0	13	0	7
InstructBLIP-Vicuna-7B	2186	91	23	30	1	3	12	0	5	5	2	4	16	1	0	4	0	2	13	0	2
LLaVA-Next-8B	60	253	7	643	73	9	21	522	14	13	130	33	289	20	46	0	13	204	0	50	0

Table 6

Results of *Popular* subset on POPE metrics.

Layer	-1	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	20	21
LLaVA-1.5-7B	2057	53	59	2	2	0	4	53	0	34	0	83	0	0	0	1	1	0	50	0	1
InstructBLIP-Vicuna-7B	1286	0	8	1077	0	2	3	1	8	0	0	3	19	0	0	0	0	0	11	0	0
LLaVA-Next-8B	60	253	9	544	106	9	29	516	32	64	224	18	319	21	46	0	13	99	0	38	0

Table 7

Results of *Adversarial* subset on POPE metrics.

Layer	-1	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
LLaVA-1.5-7B	2018	76	75	1	2	0	4	53	0	33	0	84	0	0	0	1	2	0	50	0	0	0	1
InstructBLIP-Vicuna-7B	2005	92	32	29	6	4	107	0	8	5	1	4	62	1	2	4	20	1	13	1	0	0	3
LLaVA-Next-8B	60	249	22	556	119	12	34	529	36	91	244	22	222	20	45	0	13	85	0	0	2	39	0

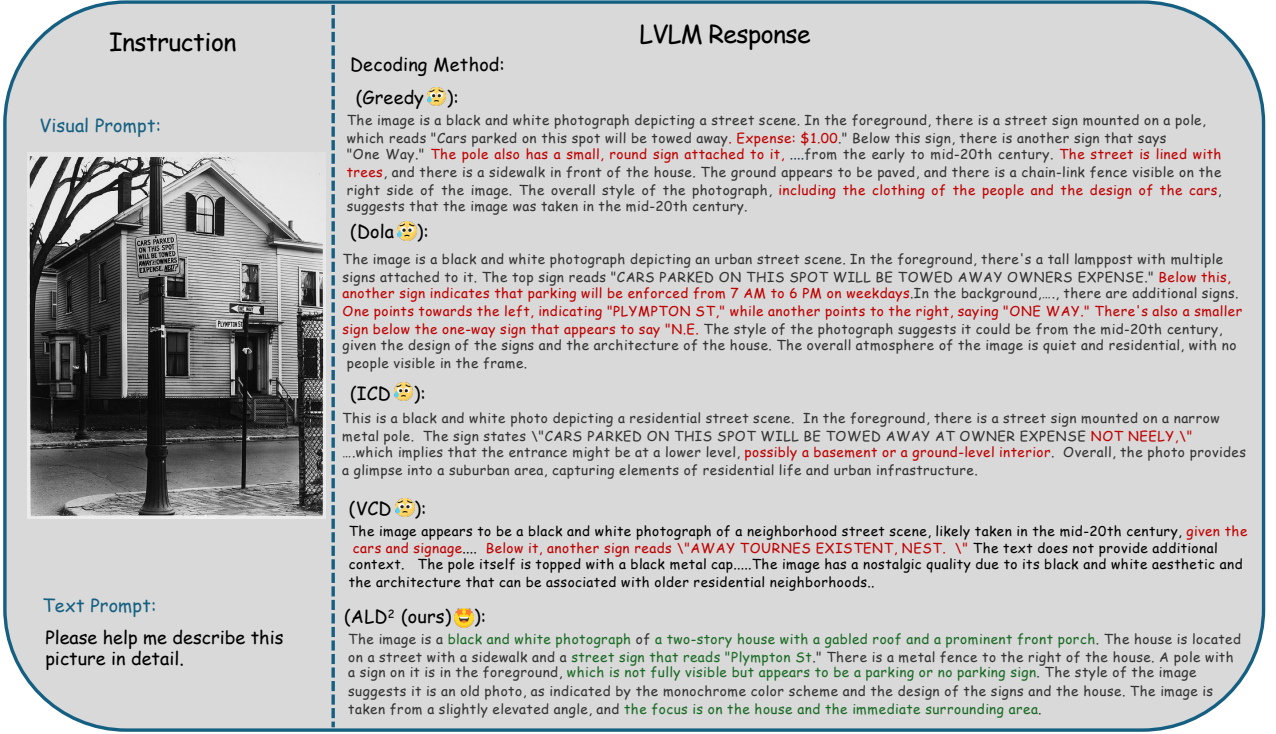


Figure 12: One case for LLaVA-Next-8B based on five decoding methods when evaluating on CHAIR metrics. The red fonts represent the hallucination words, and the green fonts are the opposite. This case illustrates all the hallucinations across five methods. The red fonts represent the hallucination words, while the green fonts are the opposite.

E. A Description about Category

The LVLMM consists of an embedding layer followed by M transformer decoder layers, indexed from 1 to M . We adopt a quantification strategy to identify the most informative layer l within the first three quarters of the candidate early-exit layers. Taking LLaVA-1.5-7B as an example, the model contains 32 decoder layers, and we consider the embedding layer together with layers 1 through 24. Thus, the search space for l is $[0, 24]$, where 0 denotes the embedding layer.

However, we also observe scenarios in which the final layer already exhibits sufficiently high confidence to decode the correct token. In such cases, we designate $l = -1$. Although this is functionally equivalent to selecting the final layer, the notation highlights that no denoising operation is applied, preventing potential misunderstanding about how the layer M could be contrasted with itself, given that l specifically denotes the denoised layer. The use of $l = -1$ explicitly marks this special case.

Consequently, the complete label space for l becomes $[-1, 24]$, yielding 26 distinct categories that the predictor is required to classify.

F. Case Study

To further validate the effectiveness of the Adaptive Layer-wise Denoising Decoding (ALD²) method, we present additional qualitative and quantitative cases across three representative Large Vision-Language Models (LVLMMs), evaluated using multiple benchmarks, including the POPE metrics, CHAIR metrics, and the MME benchmark. As illustrated in Figures 11a and 11b, ALD² consistently outperforms the four alternative decoding strategies across all tested models, demonstrating both robustness and generalizability.

A closer examination of the detailed results presented in Figure 12 reveals that ALD² maintains its advantage even under more challenging evaluation conditions, such as those posed by the CHAIR metrics, which specifically assess fine-grained object recognition and relational reasoning in complex visual scenes. In contrast, while other methods like Instruction Contrast Decoding (ICD) and Visual Contrastive Decoding (VCD) mitigate hallucinations to a certain degree, they still produce residual incorrect or irrelevant descriptions in some cases. ALD², by dynamically suppressing noisy information at each decoding layer, enables the LVLMs to eliminate hallucinations entirely, resulting in outputs that are not only more accurate but also more concise and semantically coherent.

These results highlight that ALD² does not merely reduce errors superficially but fundamentally enhances the model's ability to reason over multimodal inputs, leading to a substantial improvement in the overall reliability of LVLM-generated content. The combination of consistent performance gains across different models and metrics strongly indicates that the layer-wise denoising strategy can be a universally applicable solution for improving the trustworthiness and precision of large-scale vision-language systems.

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